Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	No antibodes detected. Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No cell lines detected Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No organisms detected Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	METHODS The protocols and informed consent forms for three clinical studies were reviewed and approved by Health Canada and by Institutional Review Board Advarra (Canada).	
Provide statement confirming informed consent obtained from study participants.	METHODS The protocols and informed consent forms for three clinical studies were reviewed and approved by Health Canada and by Institutional Review Board Advarra (Canada).	
Report on age and sex for all study participants.	**Age:**Healthy adult nonsmoking male and female volunteers, aged 18 to 50 years (study 1 and study 2) and to 60 years (study 3) with a body mass index (BMI) between 18 and 30 kg/m2 (inclusively) were eligible to participate in these studies. **Sex:**Healthy adult nonsmoking male and female volunteers, aged 18 to 50 years (study 1 and study 2) and to 60 years (study 3) with a body mass index (BMI) between 18 and 30 kg/m2 (inclusively) were eligible to participate in these studies.	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	Not detected.	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	Sample size For study 1 and study 2, a total of 64 enrolled subjects were anticipated to result in at least 60 completers, which would provide 90% power to show that the 90% confidence intervals (CIs) of the geometric mean ratios (GMRs) were contained within the acceptance range of 80.00% to 125.00%.	
Randomization	Both studies were single center, randomized, open-label, laboratory-blinded, two-treatment, two-period, two-sequence, single dose crossover studies.	
Blinding	not detected.	
inclusion/exclusion criteria	Healthy adult nonsmoking male and female volunteers, aged 18 to 50 years (study 1 and study 2) and to 60 years (study 3) with a body mass index (BMI) between 18 and 30 kg/m2 (inclusively) were eligible to participate in these studies.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Not detected.	
Define whether data describe technical or biological replicates	Not detected.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	METHODS The protocols and informed consent forms for three clinical studies were reviewed and approved by Health Canada and by Institutional Review Board Advarra (Canada).	
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	As BISO and RAMI may be administered without regard to meal and do not display a significant food effect, all three studies were conducted under fasting conditions.	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	not detected.	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	The natural logarithmic (ln) transformation of primary end points Cmax and AUC0-T for RAMI and BISO (study 1 and study 2) were used for inferential analysis using an analysis of variance (ANOVA) model, including subject effect (nested within sequence), treatment, period, as well as the sequence as fixed effect.	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	Not detected.	
If data are publicly available, provide accession number in repository or DOI or URL.	Not detected.	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	